Katherine O. Montana, M.Sc.

Department of Invertebrate Zoology, American Museum of Natural History, New York, NY
Department of Biology, City University of New York Graduate Center, New York, NY
+1 916 934 3177 | kmontana@amnh.org | OrcID: 0000-0001-8219-1119

Education

PhD Ecology, Evolutionary Biology, and Behavior, City University of New York Graduate Center *In progress*
Advisor: Dr. Jessica Ware, Curator of Invertebrate Zoology, American Museum of Natural History

MSc Integrative Biology, San Francisco State University *2023*
Thesis: A molecular phylogeny of the spider family Dictynidae, Untold stories from the archives of the Academy
Advisor: Dr. Lauren Esposito, Curator of Arachnology, California Academy of Sciences
Certificate: Graduate Opportunities to Learn Data Science, focused on biology

BA Integrative Biology, with honors, and **Anthropology,** University of California, Berkeley *2020*
Thesis: Characterizing potential tetrodotoxin resistance in Pacific Chorus Frogs associated with *Taricha* newts
Advisor: Dr. Rebecca Tarvin, Curator of Herpetology, Museum of Vertebrate Zoology

Publications

Published

8. Evangelista, D.A., Sanchez-Herrera, M., Schwartz, J., Ware, J.L., Hughes, B., Esposito, L., Banda, R., **Montana, K.O.**, Bondoc Mafla, S., Goodman, A., & Kohli, M.K. Evaluating perceptions of STEM majors to explain diversity gaps in entomology and other Sciences. *Annals of the Entomological Society of America* 119, no. 1, 25–37 (Oct 16, 2026). doi.org/10.1093/aesa/saaf035
7. **Montana, K.O.***, Gorneau, J.A.*, Gosnell, J.S., Crews, S.C., Xavier, C., Marta, K.S., Pantoja, P., Hannappel, M.P., Stiner, E.O., Jones, M.M., Mottershead, G., Cala-Riquelme, F., & Esposito, L.A. If history is written by the victors, who describes the spiders? Species author trends reflect gender and geopolitical disparities in biodiversity science. *Biological Journal of the Linnean Society* 146, no. 1, blaf067 (Sep 13, 2025). Special Issue: "Mobilizing Natural History Collections in the Global South." doi.org/10.1093/biolinnean/blaf067
6. **Montana, K.O.**, Cala Riquelme, F., Crews, S.C., Gorneau, J.A., al-Jamal, A.M., alequín, I.d., Spagna, J.C.[†], Ballarin, F., Esposito, L.A. Tailor's drawer no more: A reappraisal of the spider family Dictynidae O. Pickard-Cambridge, 1871 *sensu lato*. *Zoological Journal of the Linnean Society* 204, no. 2 (Jun 4, 2025). doi.org/10.1093/zoolinnean/zlaf007
5. **Montana, K.O.**, Gosliner, T.M., Crews, S.C., Bonomo, L.J., Carlton, J.T., Johnson, R.F. Reversing a reported case of transoceanic dispersal: Nudibranch identifications among tsunami debris. *PLOS One* 19, no. 12. (Dec 12, 2024). e0306586. doi.org/10.1371/journal.pone.0306586
4. Gorneau, J.A., Cala-Riquelme, F., Crews, S.C., **Montana, K.O.**, Spagna, J.C., Vink, C.J., Esposito, L.A. Mystery on the bounty: The family-level status of *Pacificana cockayni* Hogg, 1904 (Araneae). *Molecular Phylogenetics and Evolution* 204, 108256. (Dec 2, 2024). doi.org/10.1016/j.ympev.2024.108256
3. Gorneau, J.A., Crews, S.C., Cala-Riquelme, F., **Montana, K.O.**, Spagna, J.C., Ballarin, F., Almeida-Silva, L.M., Esposito, L.A. Webs of intrigue: museum genomics elucidate relationships of the marronoid spider clade (Araneae). *Insect Systematics and Diversity* 7, no. 5 (Sep 12, 2023). doi.org/10.1093/isd/ixad021
2. **Montana, K.O.**, Kim, R., Quesada-Khoury, K., Basas, R., Garcia, G., Richardson-Beatty, M., Voong, V., Esposito, L.A. Untold stories: Reclaiming histories of marginalized scientists from archives. *Bulletin of the Society of Systematic Biologists* 2, no. 1 (Aug 25, 2023). doi.org/10.18061/bssb.v2i1.9189
1. **Montana, K.O.**, Ramírez-Castañeda, V., Tarvin, R.D. Are Pacific Chorus Frogs (*Pseudacris regilla*) resistant to tetrodotoxin (TTX)? Characterizing potential TTX exposure and resistance in an ecological associate of Pacific Newts (*Taricha*). *Journal of Herpetology* 57, no. 2 (Jul 17, 2023): 220–28. doi.org/10.1670/22-002

In Progress

2. **Montana, K.O.**, Abbott, J.C., Abbott, K.A., Newton, L.G., Uche-Dike R., Goodman, A.M., Bybee, S.M., Guralnick, R.P., Ware, J.L. Systematics, divergence time, biogeography, and trait evolution of the dragonfly genus *Erythemis* Hagen, 1861 (Anisoptera: Odonata). *International Journal of Odonatology. In revision.*
1. Carter, P., Goodman, A.M., Abbott, J.C., Bota Sierra, C., Bybee, S.M., Dijkstra, K.D., Dow, R., Fomekong Longtchi, J., Frandsen, P., Gonzalez Mozo, L., Guralnick, R.P., Hamalainen, M., Joshi, S., Kohli, M., Lupiyaningdyah, P., **Montana, K.O.**, Newton, L.G., Onsongo, V.M., Orr, A., Pinto, A., Sanchez-Herrera, M., Sutherland, L., Theischinger, G., Ware, J.L., Zhang, H., Kalkman, V. The classification and diversity of dragonflies and damselflies (Odonata). *International Journal of Odonatology. Under review.*

*Indicates co-first-authorship

Research Talks

Oral Papers

10. **Montana, K.O.**, Kohli, M.K., Uche-Dike, R., Ware, J.W. Metabarcoding reveals Yukon River aquatic invertebrate communities via salmon guts. 2026. Aquatic Ecosystems in the Data Age Online Symposium. Royal Entomological Society.
9. **Montana, K.O.**, Kohli, M.K., Sbarro, E., Ware, J.W. Population genetics and gut microbiome of New York City cockroaches. Presented at:
    - Entomological Society of America Eastern Branch Meeting. PhD Student 10-Minute Paper Competition (1st place winner). 2026. Saratoga Springs, NY, USA.
    - Early Research Initiative Conference at the City University of New York Graduate Center. 2025. New York, NY, USA.
8. **Montana, K.O.**, Abbott, J.C., Abbott, K.A., Newton, L.G., Uche-Dike R., Goodman, A.M., Bybee, S.M., Guralnick, R.P., Ware, J.L. Systematics, divergence time, biogeography, and trait evolution of the dragonfly genus *Erythemis* Hagen, 1861 (Anisoptera: Odonata). Presented at:
    - International Congress of Odonatology. 2025. Villa de Leyva, Colombia.
    - Entomological Society of America National Meeting. PhD Student 10-Minute Paper Competition. 2025. Portland, OR, USA.
    - Entomological Society of America Eastern Branch Meeting. PhD Student 10-Minute Paper Competition. 2025. Harrisburg, PA, USA.
7. **Montana, K.O.***, Gorneau, J.A.*, da Silva Marta, K., Xavier, C., Cala Riquelme, F., alequín, I.d., Crews, S.C., Hannappel, M., Jones, M., Mottershead, G., Pantoja, P., Stiner, E., Esposito, L.A. The history of gender and geographic diversity among spider taxonomists in the Americas (1946–2021). 2024. Queer Entomology: A Retelling of Entomology Through a Queer Lens. Entomological Society of America National Meeting. Phoenix, AZ, USA.
6. **Montana, K.O.**, Cala Riquelme, F., Crews, S.C., Gorneau, J.A., Esposito, L.A. Dazzling, diverse Dictynidae: A molecular phylogeny of the spider family Dictynidae. Presented at:
    - International Congress of Arachnology. 2023. Montevideo, Uruguay.
    - Building a Better Insect Tree of Life Symposium. Entomological Society of America National Meeting. 2023. National Harbor, MD, USA.
4. **Montana, K.O.**, Richardson-Beatty, M., Kim, R., Esposito, L.A. Unsung arachnology pioneers: Harriet Exline Frizzell and the women of arachnology. 2022. American Arachnological Society, Davis, CA, USA.
3. **Montana, K.O.**, Kim, R. The problematic legacy of David Starr Jordan. 2022. Anti-Racism Seminar Series, California Academy of Sciences, San Francisco, CA, USA.
2. **Montana, K.O.**, Ramírez-Castañeda, V. Tarvin, R.D. Characterizing potential tetrodotoxin resistance in Pacific Chorus Frogs, *Pseudacris regilla*. 2020. Integrative Biology Honors Thesis Presentations (Online), UC Berkeley, Berkeley, CA, USA.
1. **Montana, K.O.**, Gosliner, T., Johnson, R.F. Why Didn't the Nudibranch Cross the Ocean? Understanding biogeographic and genetic relationships of *Hermissenda*. 2019. Summer Systematics Institute Symposium, California Academy of Sciences, San Francisco, CA, USA.

Posters

4. **Montana, K.O.**, Cala Riquelme, F., Crews, S.C., Gorneau, J.A., Esposito, L.A. Dazzling, diverse Dictynidae: A molecular phylogeny of the spider family Dictynidae, in progress. 2023. ARCS Annual Scholarship Dinner and Symposium, San Francisco, CA. USA.
3. **Montana, K.O.**, Kim, R. Untold science histories from the archives at the California Academy of Sciences. 2022. ARCS Annual Scholarship Dinner and Symposium, San Francisco, CA, USA.
2. **Montana, K.O.**, Ramírez-Castañeda, V., Tarvin, R.D. Characterizing potential tetrodotoxin resistance in Pacific Chorus Frogs, *Pseudacris regilla*. 2020. SACNAS National Diversity in STEM Conference (Online).
1. **Montana, K.O.**, Gosliner, T., Johnson, R.F. Why didn't the nudibranch cross the ocean? Understanding biogeographic and genetic relationships of *Hermissenda*. 2019. SACNAS National Diversity in STEM Conference, Honolulu, HI, USA.

Science & Cultural Communication

Invited Workshops & Public Talks

14. **Montana, K.O.** The sex lives of dragonflies. 2026. Biology on Tap. New York, NY, USA.
13. **Montana, K.O.** New York City cockroaches. 2026. American Museum of Natural History Science and Nature Program for 4th Graders. New York, NY, USA.

12. Ware, J.W., Beatty, C., Beatty, T., Gamett, E., Gonzalez Mozo, L., Harding, K., Hulick, P., Krichilsky, R., **Montana, K.O.**, Onsongo, V.M., Salas, R. Insects of American Samoa and beyond. 2026. American Samoa Community College. Malaeimi, American Samoa.
11. **Montana, K.O.** My journey as an entomologist. 2024 and 2025. Sutter Middle School STEM Academy. Folsom, CA, USA.
10. **Montana, K.O.,** Eichert, A., Gamett, E. Insectarium course entomologists panel. 2024. American Museum of Natural History. New York, NY, USA.
9. **Montana, K.O.,** Hughes, C., Raab, R., Baldauf, L., Misra, R. Artificial intelligence: Social and philosophical issues panel. 2023. ARCS Annual Scholarship Dinner and Symposium, San Francisco, CA, USA.
8. **Montana, K.O.** How I catch 'em all: Arachnology fieldwork. 2023. Pokémon NightLife, California Academy of Sciences, San Francisco, CA, USA.
7. **Montana, K.O.**, Gorneau, J.A. Arachnophilia: Using museums to understand and conserve arachnids. 2023. Marin Science Seminar, Terra Linda High School, San Rafael, CA, USA.
6. **Montana, K.O.**, Gorneau, J.A. R resources for biologists. 2022. Summer Systematics Institute Lecture Series, California Academy of Sciences, San Francisco, CA.
5. **Montana, K.O.** Uncovering untold stories: Spiders and scientists at the California Academy of Sciences. 2022. Science Action Club Educator Training (Online), California Academy of Sciences, San Francisco, CA, USA.
4. **Montana, K.O.**, Acris, K.D.P., Stark, R., Gaines, J., Heater, S., Muñoz, J. How do we know what works in science education? Panel. 2022. ARCS Annual Scholarship Dinner and Symposium, San Francisco, CA, USA.
3. Kim, R., Garcia, C., Esteves, F., **Montana, K.O.**, Palaima, L. Colonial and racist legacies at the California Academy of Sciences. 2022. Mosaic Collaborative DEIA Workshop Series (Online), California Academy of Sciences, San Francisco, CA, USA.
2. **Montana, K.O.**, Kim, R. Overshadowed women in the history of the California Academy of Sciences. 2022. Women in Science NightLife, California Academy of Sciences, San Francisco, CA.
1. **Montana, K.O**. Journey of the nudibranch, the frog, and the scientist. 2020. Summer Systematics Institute Lecture Series (Online), California Academy of Sciences, San Francisco, CA.

Media & Popular Articles

9. Ware, J.W., **Montana, K.O.**, Millena, R.J.A, Eichert, A., Gamett, E., Krichilsky, E. Opinion: Trump's war on diversity is having a devastating impact on critical scientific fields. 2025. New Jersey Star-Ledger.
8. Gorneau, J.A., **Montana, K.O.**, Esposito, L.A. 2023. Marronoids: The (mostly) little brown spiders with a big story to tell. Entomology Today.
7. **Montana, K.O.** What's in a name? 2023. KQED Perspectives.
6. **Montana, K.O.**, Kim, R., Quesada-Khoury, K., Basas, R., Garcia, G., Richardson-Beatty, M., Voong, V. Untold stories from the archives. 2022. California Academy of Sciences Website, San Francisco, CA.
5. **Montana, K.O.**, Kim, R. The problematic legacy of David Starr Jordan. 2022. California Academy of Sciences Website, San Francisco, CA.
4. **Montana, K.O**. From the shore to the redwoods: Celebrating forty years of biodiversity. 2021. Native Animal Rescue of Santa Cruz County Blog, Santa Cruz, CA.
3. **Montana, K.O.** What's in a name? An introduction by Katherine Montana. 2021. MixedLife.net.
2. Tarvin Lab, Amphibioscope. 2020.
1. **Montana, K.O**. *Boophis ankarafensis.* 2020. AmphibiaWeb, University of California, Berkeley, CA.

Grants, Fellowships & Honors

Total received for all categories: $126,251

Research and Travel Grants

2026	**$4,000**	Pathfinder Grant	*The Explorers Club*
2026	**$1,500**	Doctoral Student Research Grant	*City University of New York Graduate Center*
2026	**$1,400**	Black & Schwartz Science Conference Travel Award	*City University of New York Graduate Center*
2025	**$8,200**	David Redden Conservation Science Fund Small Grant for Conservation Research	*Black Rock Forest*
2025	**$4,000**	Connect New York Summer Research Fellowship	*City University of New York Graduate Center*
2025	**$3,000**	Graduate Student Research Award	*Society of Systematic Biologists*
2025	**$2,500**	Maxwell/Hanrahan Grant	*American Museum of Natural History*
2025	**$2,350**	Theodore Roosevelt Memorial Grant	*American Museum of Natural History*
2025	**$2,859**	LinnéSys: Systematics Research Fund ($2,059 in 2025 & $800 in 2022)	*Linnean Society of London*
2025	**$1,750**	Sydney Anderson Travel Award	*American Museum of Natural History*
2025	**$1,600**	Southeast Asia Consortium Field Research Grant	*State Univ. of NY and City Univ. of New York*
2023	**$1,000**	ESA Program Enhancement Travel Fund	*Entomological Society of America*
2022	**$400**	Instructionally Related Activities (IRA) Student Research Award	*San Francisco State University*

2019 **$2,000** Bio Research Experience for Undergraduates Travel Grant *National Science Foundation*

Fellowships & Scholarships

2024 **$52,500** Richard Gilder Graduate School Fellowship *American Museum of Natural History*
2022 **$20,000** Achievement Reward for College Scientists Scholarship ($10K 2021 & $10K 2022) *ARCS Foundation*
2022 **$14,352** Graduate State University Grant ($7,176 2021 & $7,176 2022) *San Francisco State University*
2022 **$2,000** Women in Science and Engineering Award *San Francisco State University*
2021 **$300** Broadening Participation Scholarship *Society for Integrative and Comparative Biology*
2020 **$165** Registration Scholarship *SACNAS National Diversity in STEM Conference*

Awards

2026 **$100** PhD Presentation Competition – 1st Place *Eastern Branch Meeting, Entomological Society of America*
2020 **$250** Natural History Award *Department of Integrative Biology, UC Berkeley*
2019 **$125** Outstanding Research Presentation *SACNAS National Diversity in STEM Conference*

Experience

Research & Collections Appointments

PhD Student Researcher, Dept. of Invertebrate Zoology *Aug 2024–present*
American Museum of Natural History, New York, NY
Mentor: Dr. Jessica Ware

Master's Student Researcher, Dept. of Entomology *Aug 2021–Jun 2024*
Institute for Biodiversity Science and Sustainability, California Academy of Sciences, San Francisco, CA
Mentor: Dr. Lauren Esposito

Library Research Assistant *Sep 2021–Jun 2022*
California Academy of Sciences, San Francisco, CA
Mentor: Rebekah Kim, MLS

Undergraduate Honors Researcher, Museum of Vertebrate Zoology *Aug 2019–Aug 2020*
University of California, Berkeley, Berkeley, CA
Mentor: Dr. Rebecca Tarvin

Summer Systematics Institute Intern *Jun 2019–Aug 2019*
NSF Research Experience for Undergraduates, California Academy of Sciences, San Francisco, CA
Mentor: Drs. Rebecca Johnson and Terrence Gosliner

Collections Assistant *Aug 2018–May 2019*
Phoebe A. Hearst Museum of Anthropology, University of California, Berkeley, Berkeley, CA
Mentor: Leslie Freund, MA

Undergraduate Research Apprentice, Dept. of Environmental Science, Policy, and Management *Aug–Dec 2018*
University of California, Berkeley, Berkeley, CA
Mentor: Dr. David Kurz

Mentorship, Teaching & Outreach

Adjunct Lecturer & **Graduate Teaching Assistant,** Ecology & Evolution (BIO 228) *Aug–Dec 2024*
Dept. of Biology, City College of New York, New York, NY
Professor: Dr. Robert Anderson

Summer Research Mentor, Dept. of Invertebrate Zoology *Jun–Jul 2025*
American Museum of Natural History, New York, NY
Mentee: Ethan Sbarro

Explainer: Facilitator, Exploratorium *Aug 2023–Jul 2024*
San Francisco, CA

Research Mentor and **Co-coordinator** of the **Summer Systematics Institute** *Jun–Jul 2022, 2023*
Institute for Biodiversity Science and Sustainability, California Academy of Sciences, San Francisco, CA
Mentee: David Canseco

Library Research Mentor *Jan–May 2022*
Academy Library, California Academy of Sciences, San Francisco, CA
Mentees: Cecilia Alvarado, Racqyl Basas, Gabriella Garcia, Karim Quesada-Khoury, Marion Richardson-Beatty, Viva Voong

Program Assistant and **Cohort Member** of the **Outdoor Educators Institute** *Feb 2021–Nov 2021*
Justice Outside, Oakland, CA

Undergraduate Student Instructor, IB104 Natural History of the Vertebrates *Jan–May 2020*
Dept. of Integrative Biology, University of California, Berkeley, Berkeley, CA

Professors: Drs. Rauri Bowie, Jim McGuire, Alan Shabel

Societies & Service

Partner University Student Rep for the **Comparative Biology PhD Program Committee** *Nov 2025–present*
City University of New York & American Museum of Natural History
Social Media Coordinator for the **Worldwide Dragonfly Association** *Aug 2025–present*
Treasurer's Assistant for the **Dragonfly Society of the Americas** *Feb 2025–present*
Administrative Assistant of **EntoPOC (Entomologists of Color)** *Mar–Jun 2024*
Graduate Student Representative for **Curators' Forum** *Sep 2022–Jun 2023*
Institute for Biodiversity Science and Sustainability, California Academy of Sciences
Society for the Study of Evolution member *2025–present*
Entomological Society of America member *2022–present*
Member of Larvae-MNH, Entomology Games team, 2nd place at regionals 2026 *2025–present*
Society for Systematic Biologists member *2021–present (intermittent)*

Symposia organized

Brown, V., Ware, J.L., **Montana, K.O.**, Walker, A.E. 2025. Radical Metamorphosis: Queer Entomology as a Force for Change. Entomological Society of America National Meeting. Portland, OR, USA.

Fieldwork

Tutuila, American Samoa (all insects) *Jan 2026*
Black Rock Forest, NY, USA (dragonflies) *Jun 2025*
Karanambu, Guyana (all insects) *Jan 2025*
Crete, Greece (spiders) *Apr 2022*
Georgia, Alabama, Mississippi, Louisiana, USA (spiders) *Mar 2022*
Northern California, USA (various sites, scorpions) *Aug 2021*
Briones Regional Park, Martinez, CA, USA (frogs, newts) *~Monthly Oct 2019–Oct 2020*